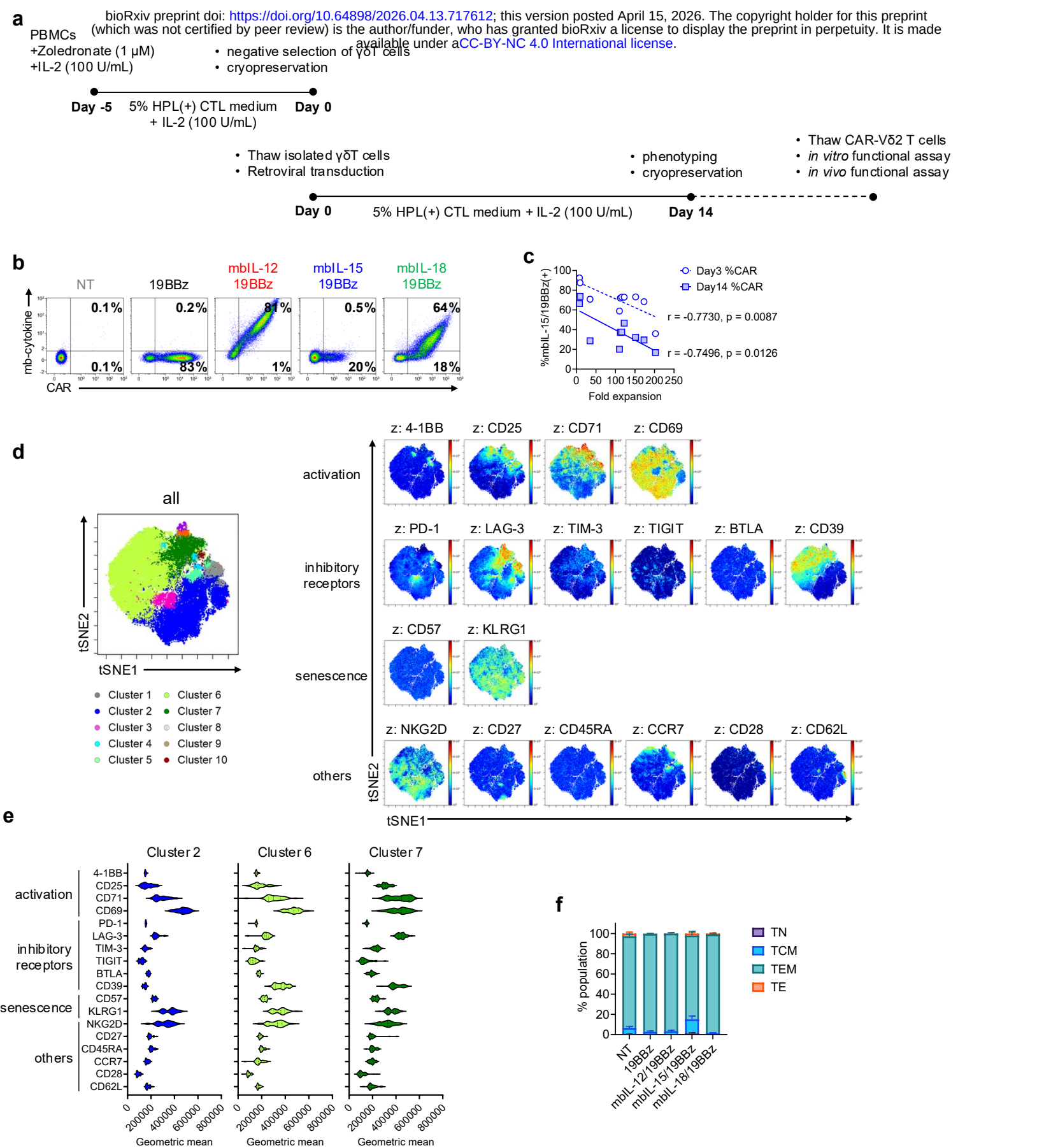
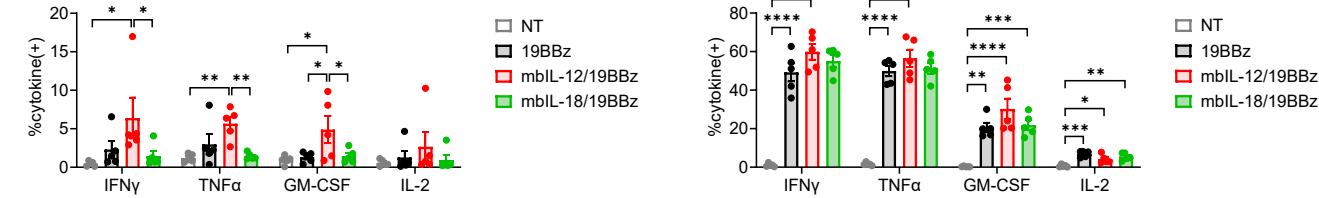




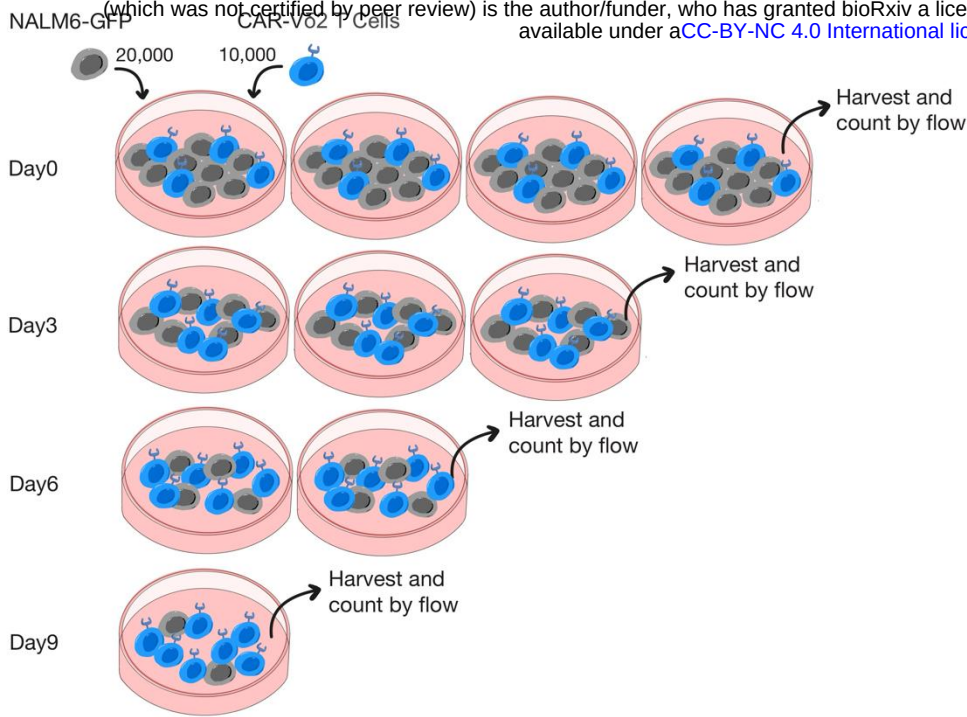
Supplementary Figure 1. Additional surface phenotype of gene-modified V δ 2 T cells.

a, Schematic of gene modification and manufacturing of V δ 2 T cells used in this study. **b**, Representative flow plots of surface expression of CAR and mbCyto on day 14 post transduction from 11 independent donors. **c**, Graph showing correlation of % mbIL-15/19BBz(+) cells and total fold cell expansion during 14-day manufacturing. **d**, Visualization of t-distributed stochastic neighbor embedding plots (visNE) of concatenated events from 5 different cell products with 18 surface phenotype markers followed by FlowSOM metaclustering (cluster 1-10). Left visNE plot showing all concatenated events including all V δ 2 T cell products with clustering, and right visNE plots showing overlay of the geometric mean of individual surface marker expression. **e**, Violin plot showing geometric mean of all surface markers expressed in cluster 2, 6 and 7 (mean $\pm$ S.E. n=8). **f**, Bar graph summarizing memory phenotype of each V δ 2 T cell product based on CD27 and CD45RA expression (TN: CD27+CD45RA+, TCM: CD27+CD45RA-, TEM: CD27-CD45RA-, TE: CD27-CD45RA+).



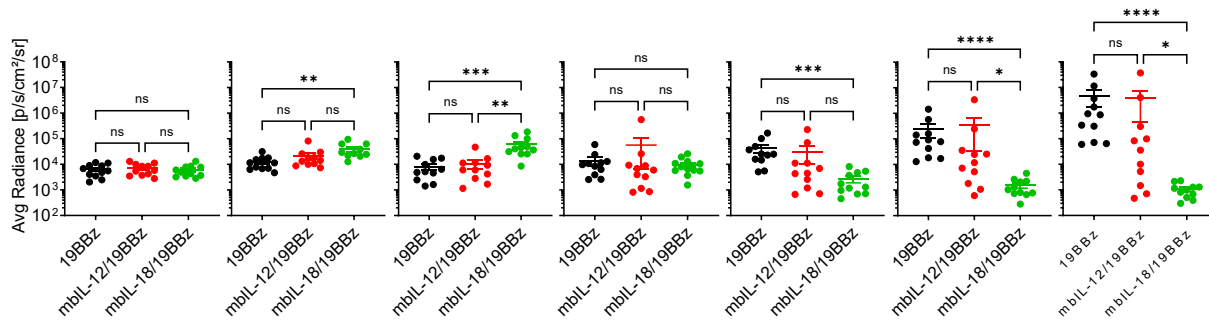
Supplementary Figure 2. Individual cytokine expression with intracellular cytokine staining.

Results of intracellular cytokine staining in each V δ 2 T cell product without antigen stimulation (**a**) or with antigen stimulation (**b**) for 4 hrs (mean $\pm$ S.E. n=5). Statistical differences are calculated by One-way ANOVA with Tukey's multiple comparison in each cytokine (a, b). *p $\leq$ 0.05, **p $\leq$ 0.01, ***p $\leq$ 0.001, ****p $\leq$ 0.0001.



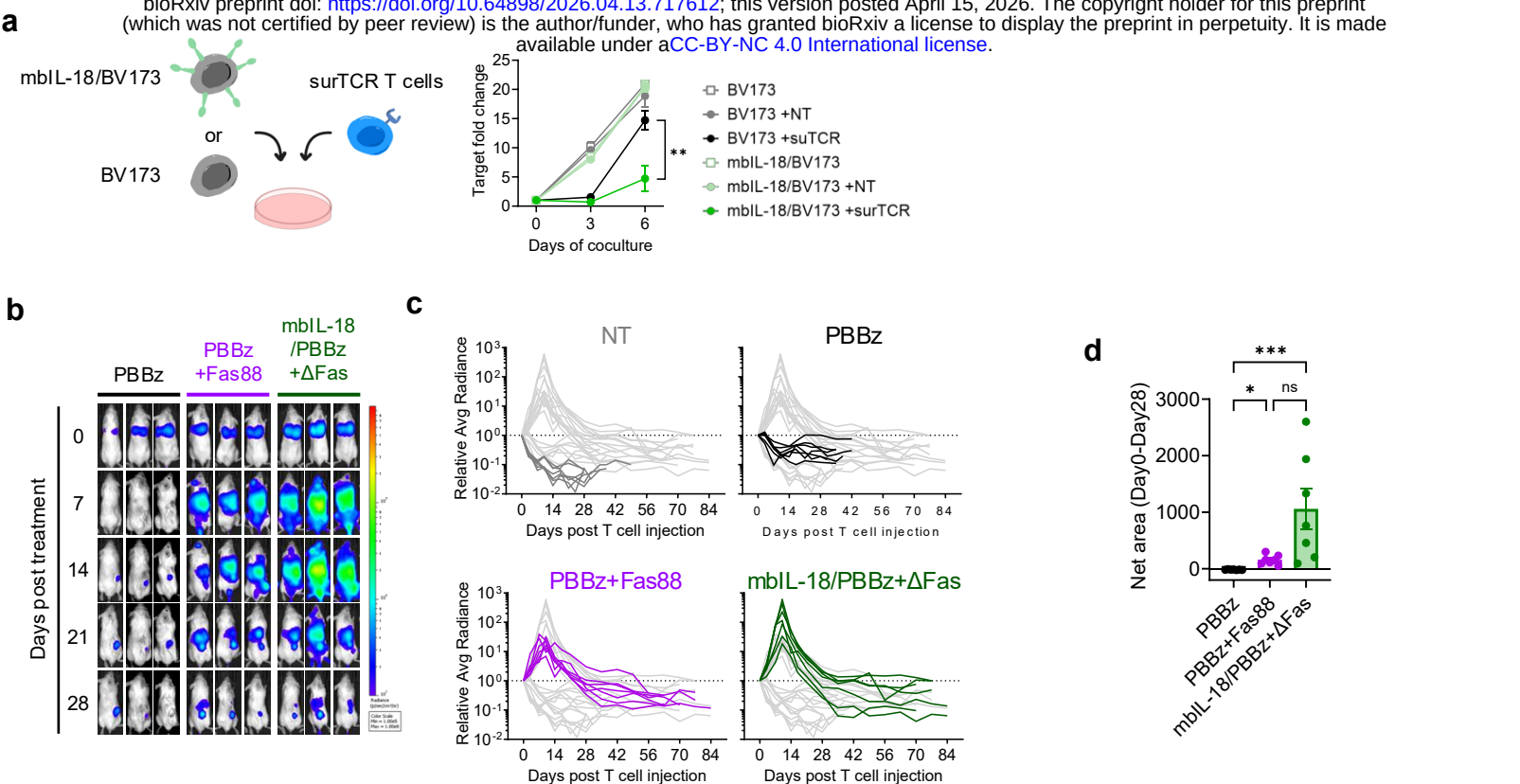
Supplementary Figure 3. Schematic of *in vitro* 9-day coculture set-up used in this study.

Effector cells and target cells are cocultured in indicated effector to target ratio. On the day of set up, total 4 wells are prepared per condition, and every 3 days, cells are harvested from one well and subjected for cell counting on flow cytometry while half of culture medium are replaced for remaining wells for next time point.



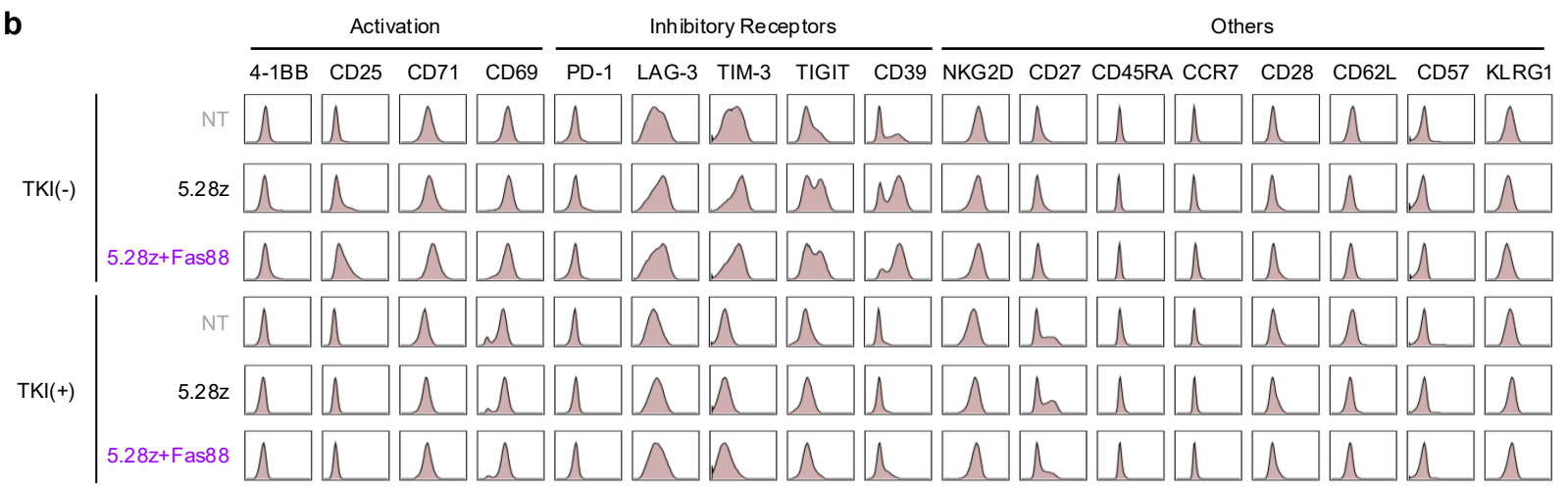
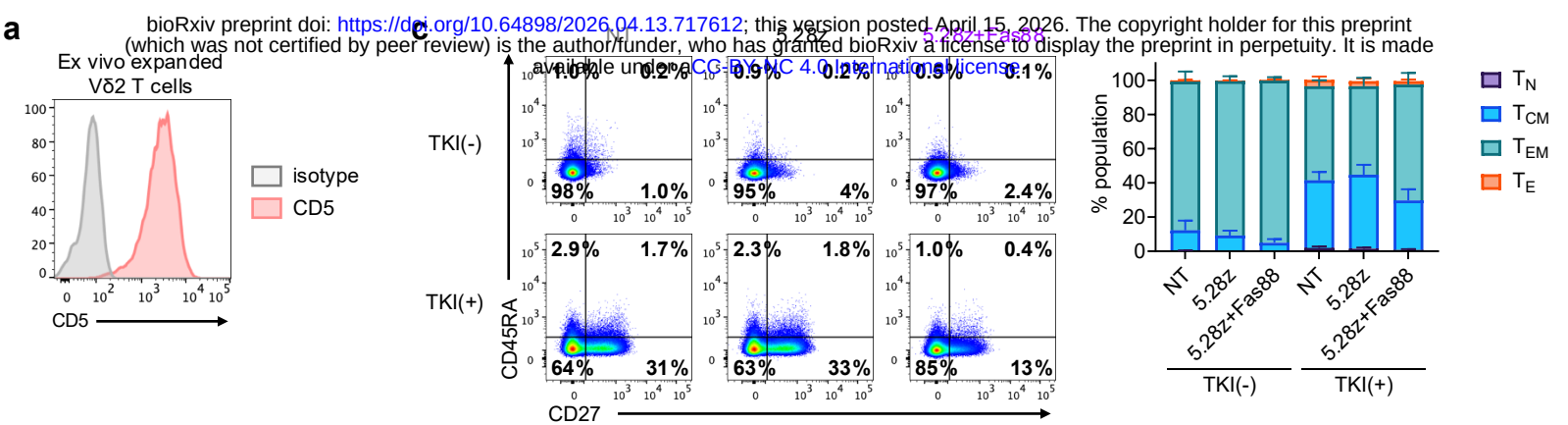
Supplementary Figure 4. mbIL-18/19BBz V52 T cells delay *in vivo* tumor killing at early time point.

Graphs summarizing tumor bioluminescence from individual mice in each treatment group (n=11 in 19BBz and mbIL-12/19BBz, n=12 in mbIL-18/19BBz). Statistical differences are calculated by One-way ANOVA with Tukey's multiple comparison. *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001. ns; non-significant.



Supplementary Figure 5. Effect of mbIL-18 trans-presentation on α T cells and V52 T cell bioluminescence in CFPAC-1 xenograft mouse model.

a, Schematic of coculture experiments of survivin-specific TCR transduced T cells against BV173 cells with or without mbIL-18 modification (left) and line graph showing target fold change during 6-day coculture (mean $\pm$ S.E. n=3). **b**, Representative mice images of V52 T cell bioluminescence at the indicated time points. **c**, V52 T cell expansion and persistence in individual mice over time (n=6 in no treatment and 19BBz, n=5 in 19BBz+Fas88 and mbIL-18/19BBz+ΔFas). **d**, Bar graph showing Net area (baseline = 1) of AkaLuc bioluminescence fold change from day 0 to day 28. Statistical differences are calculated by Paired t test at day 6 (a), One-way ANOVA with Tukey's multiple comparison (d). *p $\leq$ 0.05, **p $\leq$ 0.01, ***p $\leq$ 0.001, ns; non-significant.



Supplementary Figure 6. Additional phenotype of CD5.CAR Vδ2 T cells

a, Histogram of CD5 surface expression in Vδ2 T cells on the day of retroviral transduction (day 5 of manufacturing process). **b**, Histograms of individual surface marker expression in each of the 6 conditions from visualization of t-distributed stochastic neighbor embedding (viSNE) analysis. **c**, Representative flow plots of the memory phenotype of Vδ2 T cells based on CD27 and CD45RA surface expression (left) and stacked bar graph summarizing memory phenotype in 4 donors/condition (TN: CD27+CD45RA+, TCM: CD27+CD45RA-, TEM: CD27-CD45RA-, TE: CD27-CD45RA+).

Supplementary Table 1: List of antibodies used in this study

Marker	Fluorochrome	clone	Product #	Vendor
2B4	BV421	2-69	393504	BioLegend
4-1BB	BV711	4B4-1	309832	BioLegend
BTLA	APC/Cy7	MIH26	344518	BioLegend
CCR7	FITC	150503	561271	BD Biosciences
CD103	PerCP/Cy5.5	Ber-ACT8	350226	BioLegend
CD16	BV570	3G8	302036	BioLegend
CD25	PE	M-A251	555432	BD Biosciences
CD27	PE/Cy7	1A4CD27	A54823	Beckman Coulter
CD271 (NGFR)	BV421	C40-1457	562562	BD Biosciences
CD271 (NGFR)	BV510	C40-1457	563451	BD Biosciences
CD271 (NGFR)	Pacific Blue	ME20.4	345132	BioLegend
CD28	BV750	CD28.2	302970	BioLegend
CD3	APC/Fire810	SK7	344858	BioLegend
CD3	cFluor R685	SK7	R7-20345	CytekBio
CD3	APC/Alexa Fluor750	UCHT1	A66329	Beckman Coulter
CD39	BV480	TU66	746454	BD Biosciences
CD45	APC/Fire810	HI30	304076	BioLegend
CD45RA	Alexa Fluor 532	HI100	58-0458-42	Thermo Fischer
CD56	APC Fire 810	HCD56	318356	BioLegend
CD57	BV605	QA17A04	393304	BioLegend
CD62L	BV785	GREG-56	304830	BioLegend
CD69	ECD	TP1.55.3	6607110	Beckman Coulter
CD71	PE/Fire 640	CY1G4	334142	BioLegend
Fas	PE	DX2	555674	BD Biosciences
Fas ligand	PE/Cy7	NOK-1	306418	BioLegend
FLAG (DYKDDDDK)	BV421	L5	637322	BioLegend
FLAG (DYKDDDDK)	PE/Cy7	L5	637324	BioLegend
G4S Linker	Pacific Blue	E7O2V	44962S	Cell Signaling Technologies
G4S Linker	Alexa Fluor 647	E7O2V	69782L	Cell Signaling Technologies
GM-CSF	PE/Dazzle 594	BVD2-21C11	502318	BioLegend
IFN γ	V450	B27	560371	BD Biosciences
IL-12	Biotin		BAF219	R&D System
IL-15	Biotin	BH1543	515103	BioLegend
IL-18	Biotin	159-12B	D045-6	MBL Life Science
IL-2	APC/Fire750	MQ1-17H12	500352	BioLegend
KLRG1	PerCP eFluor 710	13F12F2	46-9488-42	Thermo Fischer
LAG-3	PE-Cy5	11C3C65	369346	BioLegend
NKG2D	APC	1D11	320808	BioLegend
PD-1	BV650	EH12.2H7	329950	BioLegend
Phospho-p38 MAPK (pT180/pY182)	Alexa Fluor 647	36/p38	612595	BD Biosciences
Phospho-STAT4 (pY693)	Alexa Fluor 647	38/p-Stat4	558137	BD Biosciences
Phospho-STAT5 (Tyr694)	PE-Cy7	A17016B.Rec	936908	BioLegend
Streptavidin	APC		554067	BD Bioscience
TIGIT	PE-Fire 810	A15153G	372745	BioLegend
TIM-3	Alexa Fluor 700	F38-2E2	56-3109-42	Thermo Fischer
TNF α	PE-Cy7	MAb11	557647	BD Biosciences
V δ 1TCR	VioBlue	REA173 R9.12	130-120-443	Milteny Biotec
V δ 1TCR	Biotin	REA173 R9.12	130-120-439	Milteny Biotec
V δ 2TCR	BV510	B6	331432	BioLegend
V δ 2TCR	PE	B6	555739	BD Biosciences
V δ 2TCR	Biotin	REA771 123R3	130-111-008	Milteny Biotec
$\alpha\beta$ TCR	Biotin	REA652 BW242/412	130-113-537	Milteny Biotec
$\alpha\beta$ TCR	FITC	WT31	347773	BD Biosciences
Phospho-NF- κ B p65 (Ser536)		93H1	#3033	Cell Signaling Technologies
Phospho-IKK α / β (Ser176/180)		16A6	#2697	Cell Signaling Technologies
Phospho-p38 MAPK (Thr180/Tyr182)			#9211	Cell Signaling Technologies
GAPDH		6C5	sc-32233	Santa Cruz
Goat anti-Rabbit IgG Secondary Antibody	IRDye® 800CW		926-32211	LICORBio
Goat Anti-Mouse IgG Secondary Antibody	IRDye® 680RD		926-68070	LICORBio